

# Midterm Review

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# Today's Objectives

- **Review** key course content:
  - ML & AI in Clinical Bioinformatics
  - Data Preprocessing & Cleaning
  - Supervised Learning
  - Unsupervised Learning
- **Sample Questions**



Review (so far)

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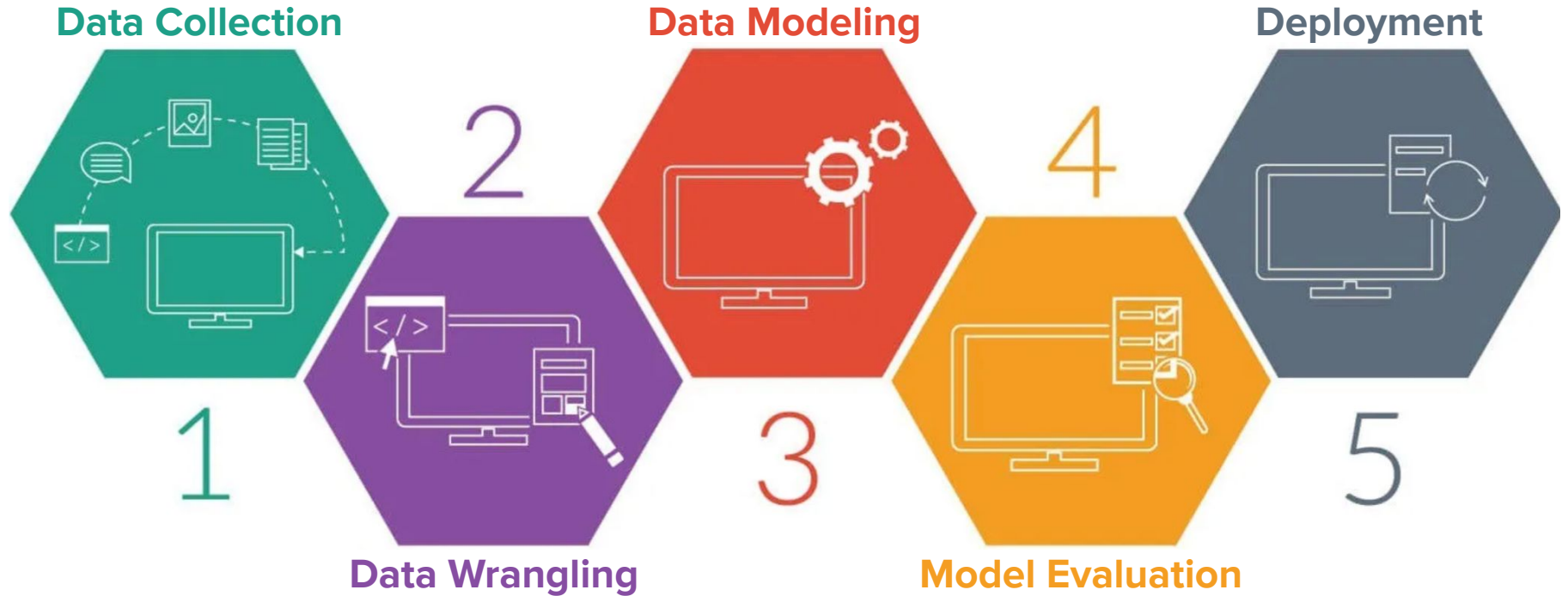
# Essential Terminology

| Terminology | Definition                                                                                        | Example(s)                                                                  |
|-------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Feature     | Property or characteristic of the data used for analysis or modeling                              | Age, blood pressure, gene expression levels                                 |
| Label       | Target variable or outcome that a model is designed to predict                                    | Survival, gene expression, disease state                                    |
| Algorithm   | A set of mathematical instructions or procedures used to create a ML model                        | Linear regression, decision trees, k-means clustering                       |
| Model       | Output of a ML algorithm after it has been trained on data; used to make predictions or decisions | Logistic regression model predicting disease risk based on patient features |

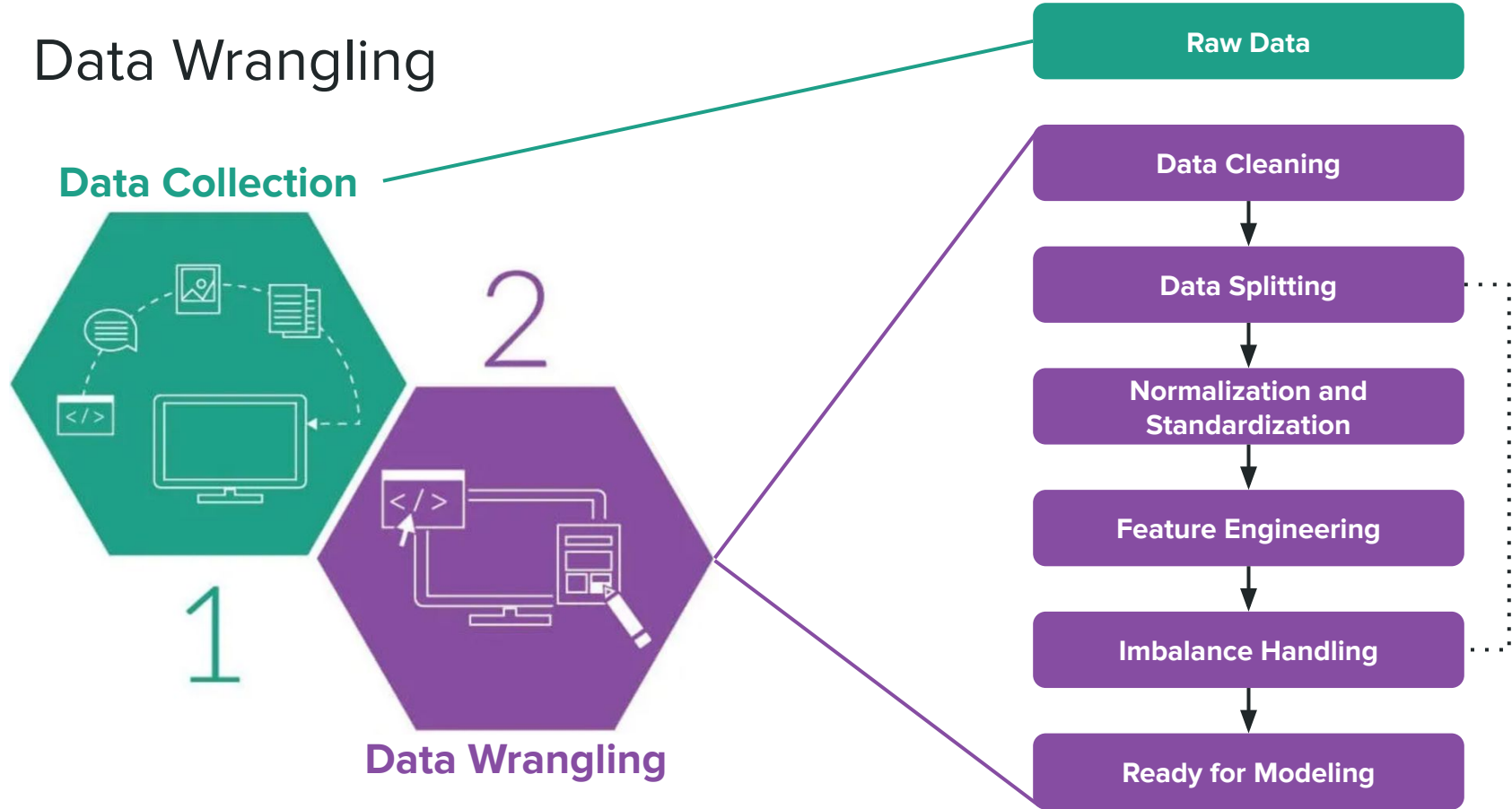
## Essential Terminology (cont'd)

| Terminology    | Definition                                                                                                               | Example(s)                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Data splitting | Dividing the data into different sets for training and optimization/evaluation                                           | 80% of data for training a model, 20% of data for testing a model                                 |
| Overfitting    | A situation where a model learns the noise or patterns specific to the training data rather than generalizable patterns. | A model that performs perfectly on training data but fails to predict accurately on testing data. |
| Regularization | Techniques used to prevent overfitting by penalizing overly complex models.                                              | Adding a penalty term to the loss function in linear regression (L1 or L2 regularization).        |

# ML & AI in Clinical Bioinformatics



# Data Wrangling



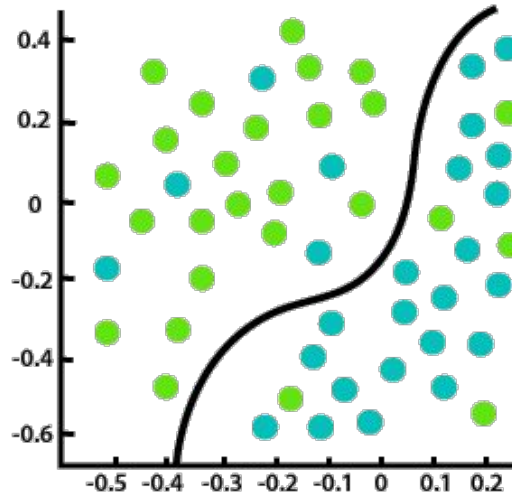
# Supervised Learning

- **Regression:**

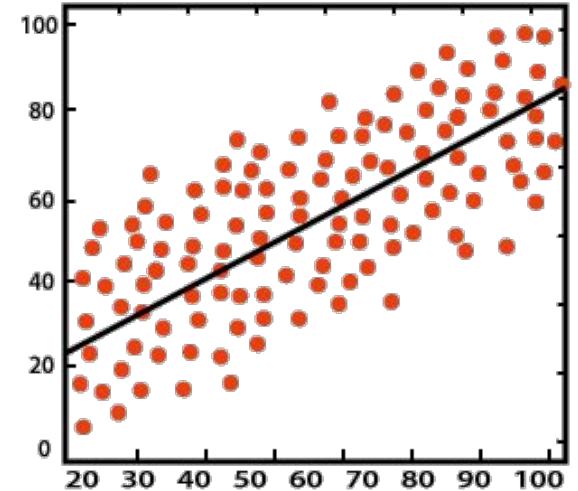
- Linear
- Polynomial

- **Classification:**

- Logistic regression
- k-Nearest Neighbours
- Decision Trees
- Random Forests
- SVM



Classification



Regression



# Regression Models

| Category         | Type/Metric           | When to Use                                                                                         | When Not to Use                                                                |
|------------------|-----------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Regression Types | Linear Regression     | - Predicting continuous outcomes with a linear relationship.                                        | - When data has non-linear patterns or interactions.                           |
|                  | Logistic Regression   | - Predicting binary or categorical outcomes (e.g., 0/1, true/false).                                | - When target variable is continuous or the relationship is highly non-linear. |
|                  | Polynomial Regression | - When the relationship between variables is non-linear but can be expressed with polynomial terms. | - When data is high-dimensional, as it increases overfitting risk.             |

# Regression Evaluation Metrics

| Category           | Type/Metric                    | When to Use                                                                      | When Not to Use                                                                                      |
|--------------------|--------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Evaluation Metrics | MSE (Mean Squared Error)       | - Assessing model accuracy for continuous targets (lower = better).              | - When evaluating classification tasks (doesn't apply).                                              |
|                    | RMSE (Root Mean Squared Error) | - Measuring prediction error for continuous targets (lower = better).            | - When evaluating classification tasks or comparing models where interpretability is more important. |
|                    | R-squared                      | - Measuring how well the model explains variance in continuous target variables. | - When comparing models with different numbers of predictors or in classification problems.          |

# Regularization in Regression Models

| Category               | Type/Metric | When to Use                                                                                | When Not to Use                                                                                               |
|------------------------|-------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Regularization Methods | L1 (Lasso)  | - Feature selection (shrinks less relevant coefficients to 0).                             | - When you don't want to exclude any feature, or multicollinearity is a major issue.                          |
|                        | L2 (Ridge)  | - Preventing overfitting by shrinking coefficients, works well with correlated predictors. | - When sparse solutions are needed or feature selection is important.                                         |
|                        | Elastic Net | - Combining L1 and L2 for feature selection with correlated predictors.                    | - When the simplicity of pure L1 or L2 regularization is sufficient or computational efficiency is a concern. |

# Classification Algorithms

| Algorithm      | Strengths                            | Weaknesses                           | Hyperparameters*                 |
|----------------|--------------------------------------|--------------------------------------|----------------------------------|
| KNN            | Simple, interpretable                | Sensitive to noise, requires scaling | $k$ (neighbors), distance metric |
| Decision Trees | Interpretable, fast to train         | Prone to overfitting                 | Max depth, min samples per split |
| Random Forest  | Reduces overfitting, robust          | Can be slow with many trees          | Number of trees, max features    |
| SVM            | Effective in high-dimensional spaces | Computationally expensive            | Kernel type, C, gamma            |

# Classification Evaluation Metrics

| Metric    | Definition                         | Best Use Case               |
|-----------|------------------------------------|-----------------------------|
| Accuracy  | Overall correctness                | Balanced datasets           |
| Precision | Positive predictive value          | When false positives matter |
| Recall    | Sensitivity/True positive rate     | When false negatives matter |
| F1 Score  | Precision-recall balance           | Imbalanced datasets         |
| AUROC     | Distinguishes between classes      | Overall performance         |
| AUPRC     | Focus on precision-recall tradeoff | Imbalanced datasets         |

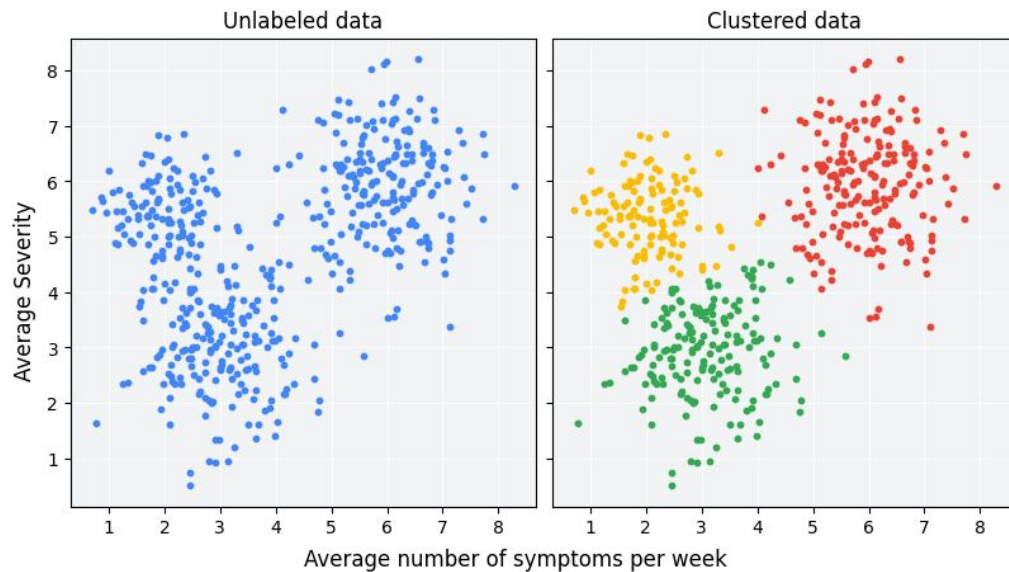
# Unsupervised Learning

- **Clustering:**

- k-Means
- Hierarchical

- **Dimensionality Reduction:**

- PCA
- UMAP
- t-SNE



# Clustering Algorithms

| Feature            | KMeans Clustering                                           | Hierarchical Clustering                                         |
|--------------------|-------------------------------------------------------------|-----------------------------------------------------------------|
| Definition         | Partitions data into $K$ clusters by optimizing centroids.  | Builds a tree-like hierarchy of clusters (dendrogram).          |
| Approach           | <b>Iterative</b> optimization.                              | <b>Agglomerative</b> (bottom-up) or <b>Divisive</b> (top-down). |
| Number of Clusters | Requires predefined $K$ .                                   | Does not require specifying $K$ beforehand.                     |
| Cluster Shape      | Assumes spherical, equal-sized clusters.                    | Works well for clusters of arbitrary shapes and sizes.          |
| Initialization     | Centroids are randomly initialized; KMeans++ improves this. | No initialization required.                                     |

## Clustering Algorithms (cont'd)

| Feature          | KMeans Clustering                                     | Hierarchical Clustering                                          |
|------------------|-------------------------------------------------------|------------------------------------------------------------------|
| Result           | Flat partition of clusters.                           | Nested clusters in a hierarchical tree structure.                |
| Interpretability | Moderate; centroids summarize each cluster.           | High; dendrogram provides detailed visual cluster relationships. |
| Strengths        | Simple, fast, and efficient for large datasets.       | No need for K, better suited for small/medium datasets.          |
| Limitations      | Requires K, sensitive to initialization and outliers. | Computationally expensive for large datasets.                    |



# Clustering – Evaluation and Challenges

| Aspect             | Evaluation Metric                                                                                                                                                                                                      | Challenges                                                                                                                 |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Cluster Quality    | - <b>Silhouette Score:</b> Measures cohesion and separation of clusters.                                                                                                                                               | - Interpreting whether high/low scores reflect true data structure;<br>- Sensitive to noise or outliers.                   |
| Number of Clusters | - <b>Elbow Method:</b> Identifies optimal k by finding the "elbow" in an inertia plot;<br>- <b>Gap Statistic:</b> Compares clustering with random data;<br>- <b>Dendrogram:</b> Visualizes hierarchical relationships. | - No single method works universally for all datasets;<br>- Ambiguity in choosing "elbow" or interpreting dendrogram cuts. |

# Dimensionality Reduction Techniques

| Characteristic | PCA                                         | t-SNE                                                  | UMAP                                                                 |
|----------------|---------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------|
| Type           | Linear dimensionality reduction             | Non-linear dimensionality reduction                    | Non-linear dimensionality reduction                                  |
| Goal           | Maximize variance                           | Preserve local pairwise similarities                   | Preserve both local and global structures                            |
| Best Used For  | Data with linear structure, noise reduction | Complex, high-dimensional data (e.g., NLP, clustering) | Complex, high-dimensional data with both local and global structures |
| Output         | Principal components                        | Low-dimensional representation (2D/3D)                 | Low-dimensional representation (2D/3D)                               |

# Dimensionality Reduction Techniques (cont'd)

| Characteristic          | PCA                                  | t-SNE                                        | UMAP                                         |
|-------------------------|--------------------------------------|----------------------------------------------|----------------------------------------------|
| Use Cases               | Noise reduction, feature extraction  | Clustering, anomaly detection, visualization | Visualization, clustering, outlier detection |
| Computational Intensity | Low                                  | High                                         | Moderate                                     |
| Interpretation          | Easy to interpret                    | Harder to interpret                          | Easier to interpret than t-SNE               |
| Preserved Information   | Variance                             | Local pairwise similarities                  | Local and global structure                   |
| Scalability             | High (works well for large datasets) | Low (slow for large datasets)                | High (faster and more scalable than t-SNE)   |

# Sample Questions

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# Question 1

**How does PCA handle variance in data?**

- A. By ignoring variance entirely.
- B. By maximizing the variance explained by each principal component.
- C. By clustering data points with similar variance.
- D. By removing features with low variance.

## Question 2

**What is the difference between normalization and standardization?**

- A. Normalization rescales data to  $[0, 1]$ , while standardization rescales to have mean 0 and standard deviation 1.
- B. Standardization rescales data to  $[0, 1]$ , while normalization rescales to mean 0 and standard deviation 1.
- C. Both refer to the same process.
- D. Neither is a data cleaning method.

## Question 3

**How does a Random Forest classifier improve upon a single Decision Tree?**

- A. By using a single large tree instead of multiple trees.
- B. By combining predictions from multiple trees to reduce overfitting.
- C. By using k-nearest neighbors within each tree.
- D. By performing dimensionality reduction before classification

## Question 4

**What does the R-squared value indicate in a regression model?**

- A. The likelihood of multiple variables being correlated.
- B. The proportion of variance in the dependent variable that can be explained by the independent variables.
- C. The performance of a classification model.
- D. The average error of predictions



**DONT BE NERVOUS**

**YOU GOT THIS**

